

Writing Piecewise Rules for Rates and Fees

Writing the rule from a price list, using it to find a charge, and working back from a charge

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- Real price lists change rate part-way through, which is exactly what a piecewise rule describes: one formula per band, each with the condition that says when it applies.
 - Write every rule in the form $C(x) = \begin{cases} \text{formula,} & \text{condition} \\ \text{formula,} & \text{condition} \end{cases}$ — the conditions are part of the answer, not decoration.
 - **The second band starts from what has already been charged.** If \$13 has been paid by 5 miles, the rest of the ride costs $13 + 1.5(m - 5)$, not $1.5m$.
 - All money answers are in dollars; state the units of the input in your rule.
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1. A taxi charges a \$3 pick-up fee plus \$2 per mile for the first 5 miles. After 5 miles, the charge is \$13 so far plus \$1.50 for each additional mile. Let m be the distance in miles.

- (a) Write $C(m)$, the fare in dollars, as a piecewise rule with both conditions.
 (b) Find the fare for a 4-mile ride, and put it on the answer line.

Answer: _____

$$C(m) = \begin{cases} 3 + 2m, & 0 \leq m \leq 5 \\ 13 + 1.5(m - 5), & m > 5 \end{cases}$$

$$C(2) = \underline{\hspace{2cm}} \quad C(9) = \underline{\hspace{2cm}}$$

(b) Find the cost of a 5-hour visit, and put it on the answer line.

Answer: _____

4. A phone plan costs \$25 a month for up to 5 gigabytes of data. Every gigabyte beyond the fifth costs an extra \$10. Let g be the data used, in gigabytes.

- (a) Write $F(g)$, the monthly charge in dollars, as a piecewise rule with both conditions.
- (b) Find the charge for a month in which 8 gigabytes are used.

Answer: _____

5. A taxi ride using the rule from Problem 1 cost \$25.

- (a) How many miles was the ride? Put the distance on the answer line.
- (b) Explain how you knew, before solving anything, that the second branch of the rule was the one to use.

Answer: _____

6. A library charges no fine for the first 3 days a book is late. From the fourth day onwards the fine is \$0.25 for each day beyond the third. Let d be the number of days late.

- (a) Write $F(d)$, the fine in dollars, as a piecewise rule with both conditions.
- (b) Find the fine on a book returned 10 days late.

Answer: _____

7. An electricity supplier charges \$0.12 per kilowatt-hour for the first 300 kilowatt-hours in a month. Beyond that, the bill is \$36 plus \$0.18 for each extra kilowatt-hour:

$$E(k) = \begin{cases} 0.12k, & 0 \leq k \leq 300 \\ 36 + 0.18(k - 300), & k > 300 \end{cases}$$

$$E(250) = \text{_____} \quad E(500) = \text{_____}$$

8. A worker is paid \$18 per hour for up to 40 hours in a week. Every hour beyond 40 is paid at \$27 per hour. Let h be the hours worked in a week.

- (a) Write $P(h)$, the weekly pay in dollars, as a piecewise rule with both conditions.
- (b) Find the pay for a 46-hour week.

Answer: _____

9. Using the pay rule from Problem 8, a worker earned \$963 in one week.

- (a) How many hours did the worker work? Put the number on the answer line.
- (b) Explain how the amount \$963 told you which branch of the rule to use before you solved anything.

Answer: _____

10. Synthesis challenge. Two phone plans charge for n minutes of calls.

Plan A: $A(n) = 30 + 0.1n$ dollars, for every number of minutes.

Plan B: \$20 a month covering the first 200 minutes, then \$0.15 for each minute beyond 200.

- (a) Write $B(n)$ as a piecewise rule with both conditions.
- (b) For calls longer than 200 minutes, find the number of minutes at which the two plans cost the same.
- (c) Explain which plan is cheaper below that number of minutes, and give evidence for your answer.

Answer: _____